

Compressed Biogas

India's on-budget decarbonisation play — mandated demand, assured pricing, three revenue legs

The problem: imports, wasted biomass, dishonest blends

Three converging failures CBG monetises simultaneously

Imported molecules

India's CGD growth rides on spot RLNG (~\$12/MMBtu, ₹0.94/MJ). Every kg of CNG demand growth is an import at the margin — price-volatile, dollar-denominated.

Wasted feedstock

Press-mud, cattle dung, paddy straw and municipal waste are burned or dumped: stubble smoke, methane emissions, and a disposal cost — negative-value feedstock waiting for a digester.

The liquid-biofuel trap

The ethanol route funds decarbonisation through a hidden consumer levy (~4% mileage at unchanged price = ₹22,700 cr/yr) and now faces distillery overcapacity (59% utilisation). CBG has neither problem: energy parity per kg, on-budget support.

- CBG converts a waste liability into a gas molecule that is legally fungible with CNG (IS 16087:2025) — sold per kg at energy parity, no mileage penalty, no volumetric pass-through
- It undercuts imported LNG outright whenever spot runs above ~\$14.8/MMBtu — the import-substitution case stands on its own

The market: demand is mandated, supply is missing

CBG Blending Obligation (CBO) on city-gas distributors — demand by decree, stepping 1% → 5%

	CBO share	CBG needed (t/yr)	Procurement value (₹ cr/yr)
FY25-26 (live)	1%	66,700	378
FY28-29	~5%	333,500	1,891

...on the 6.67-MMT CNG(T) pool alone — PNG (domestic) obligations add further demand on top

Supply is nowhere near

- SATAT envisioned 5,000 plants; a few hundred are commissioned — the obligation ramp outruns the build-out
- Each 12-TPD plant supplies ~4,000 t/yr: the 5% CBO alone needs ~85 such plants' output

Price floor + import ceiling

- SATAT assures ₹54/kg ex-plant (+5% GST); synchronization scheme injects straight into the CGD grid
- Renewable energy at ₹1.16/MJ vs ethanol's ₹2.94 — CBG is the CHEAPEST renewable molecule India buys

The policy moat: 70 circulars de-risk every leg

GOBARDhan register, ~half dated 2025-26 — the cadence is accelerating, all support on-budget and ex-plant

Offtake	SATAT ₹54/kg assured · CBO mandate · CGD grid injection	Capital	MNRE CFA (up to ₹10 cr/plant) · SBM 2.0 VGF · SASCI state capital
Evacuation	DPI scheme funds the plant-to-grid pipeline	Feedstock	Biomass machinery assistance · crop-residue channelling
Digestate	FCO Schedule VIII fertiliser status · bulk-sale rights · MDA ~₹1,500/t	Debt	RBI priority-sector lending · AIF · AHIDF
Tax & carbon	5% GST · excise exemption on blended CNG · carbon-credit eligibility	States	7 state policies + MoPNG model template (2026)

No other Indian infrastructure class has assured price + mandated demand + subsidised capex + priority debt + a regulated second product — simultaneously.

Business model: one digester, three revenue legs

Reference plant: 12 TPD CBG · 250 TPD multi-feedstock (press-mud + agri-residue) · 330 days · 90% steady utilisation

GAS — ₹19.2 cr/yr (77%)

3,564 t CBG @ SATAT ₹54/kg, injected into the CGD grid (DPI pipeline). CBO makes the buyer's demand mandatory.

FERTILISER — ₹2.2 cr/yr (9%)

14,850 t FOM under FCO Schedule VIII @ ₹1,500/t net (incl. MDA). Bulk-sale rights direct to farmers.

CARBON — ₹0.7 cr/yr (3%)

~8,900 credits/yr @ ₹800 (2.5 tCO₂e/t CBG).
MoEFCC-recognised eligibility; upside under CCTS.

- Opex ₹10.1 cr/yr (feedstock ₹700/t delivered + ₹5 cr O&M/power/manpower) → steady EBITDA ₹12.0 cr, ~54% margin
- The fertiliser leg is load-bearing: without FOM+MDA, project IRR falls 20.6% → 16.1% — digestate discipline is a core operating competency, not an afterthought
- Feedstock strategy: anchor on sugar-mill press-mud (contracted, dense, cheap) + agri-residue swing — the cooperative-mill partnership angle below

Financials: 20%+ project IRR with a policy floor under revenue

₹60 cr capex – ₹10 cr MNRE assistance = ₹50 cr net • 70:30 debt:equity @ 9.5% (priority sector) • 15-yr life

20.6%

Project IRR

32.0%

Equity IRR (levered)

₹26.6 cr

NPV @ 12%

Year 5

Payback

1.85×

Steady DSCR

Sensitivity — project IRR (%): SATAT price × delivered feedstock cost

CBG price	Feed ₹525/t (–25%)	₹700/t (base)	₹875/t (+25%)
₹48/kg	18.9	16.3	13.5
₹54/kg (SATAT)	23.1	20.6	18.0
₹60/kg	27.1	24.7	22.2

Worst cell still 13.5% — the SATAT floor + PSL debt keep the downside financeable

• Model discipline: ramp year at 60%; carbon priced conservatively (₹800/credit); no terminal value; all levers editable in cbg_pitch_model.py

Risks and mitigants

What actually kills CBG plants — and how the structure answers it

Risk	Reality	Mitigant
Feedstock supply	The #1 failure mode — plants starve, not drown	Anchor press-mud contracts with sugar mills; multi-feed digester; BAM-funded collection machinery
Digestate offtake	FOM piles up without marketing muscle	Bulk-sale authorisation + MDA ₹1,500/t + ICAR practice guide; co-marketing with fertiliser distributors
Price revision	SATAT methodology changed May-2025	Floor has only moved UP since 2018 (₹46→₹54); CBO demand decouples volume risk from price risk
Evacuation	Trucking kills margins	DPI-funded pipeline to CGD grid; site selection ≤ 10 km from network
Execution/EPC	Commissioning delays burn the interest clock	Proven-technology EPC (fixed-price), staged drawdown, MNRE CFA milestone-linked

- Deliberately NOT relying on: the NCDC coop-mill scheme as capital (96.5% goes to working capital), or carbon prices above ₹800

Landscape: majors, IPOs, and a fragmented middle

The capital markets are already pricing this theme

Majors committed

Reliance (target ~100 CBG plants), Adani TotalEnergies, GAIL, IOCL/BPCL/HPCL joint ventures — strategic buyers of both gas and, eventually, platforms

IPO pipeline live

TruAlt Bioenergy and Godavari Biorefineries listings establish public-market comps for bioenergy platforms; Praj Industries anchors the EPC value chain

Fragmented middle = the opportunity

Hundreds of single-plant developers with LOIs but no capital or digestate muscle — a roll-up / platform play with shared EPC, O&M and FOM marketing beats one-off plants

- Cooperative sugar mills are the underexploited entry: 229 functional CSMs hold the press-mud and the land, NCDC keeps them liquid, but only ₹251 cr of the ₹10,005 cr scheme went to ethanol/energy assets — they need a capital + execution partner, not another loan
- Structures that fit: BOT/BOOT on mill land, feedstock-for-equity JVs, or O&M-plus-offtake partnerships — mill gets waste disposal + lease income, platform gets contracted feedstock

The ask: seed a 10-plant CBG platform

₹150 cr

equity for 10 × 12-TPD plants

₹120 cr

EBITDA/yr at steady state

32%

levered equity IRR per plant

~40 kt/yr

CBG = 12% of the FY29 CBO gap

- Phase 1 (18 months): 3 plants on contracted sugar-mill press-mud in UP/Maharashtra — sites within 10 km of CGD networks, MNRE CFA + PSL debt term sheets in hand
- Phase 2: 7 plants replicating the template; shared EPC frame contract, central O&M, one FOM marketing organisation across the fleet
- Exit paths: strategic sale to a major building CBG portfolios, CGD backward-integration, or the public-market route the current IPO window is validating
- Every revenue leg sits on a government floor (SATAT, CBO, MDA) and every cost leg on a subsidy (CFA, DPI, PSL) — the pitch is not a technology bet, it is disciplined execution against a policy stack that is accelerating